

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Revision Test 2026 — ANSWER KEY (Version 1)

Chemistry — Class 9 (Science Group)

Syllabus: Chapter 1 – Nature of Chemistry in Science | Chapter 2 – Matter | Chapter 3 – Atomic Structure | Chapter 4 – Periodic Table & Periodicity of Properties | Chapter 5 – Chemical Bonding | Chapter 6 – Stoichiometry | Chapter 7 – Electrochemistry | Chapter 8 – Energetics | Chapter 9 – Chemical Equilibrium

Total Marks: 50

Time: 2 Hours 15 Minutes

Paper: Exam-22 (Version 1)

SECTION A — MCQ Answer Key (10 × 1 = 10 Marks)

#	Question	Correct answer	Explanation	Source in her textbook
1	The process of applying scientific knowledge to practical applications, resulting in the creation of tools, machines and systems, is called:	B. technology 1	Section 1.5 defines technology in exactly these words. Science builds and organizes knowledge; engineering uses science and mathematics to design and construct systems and structures. Technology and engineering are the confusable pair here.	Ch1 \$1.5; SLO 4 (differentiate science, technology and engineering)
2	In diamond, each carbon atom is covalently bonded to:	D. four other carbon atoms in a rigid tetrahedral network 1	Section 2.3: in diamond each carbon is bonded to four others in a tetrahedral 3-D network. Three per atom in flat hexagonal layers is graphite, the twin allotrope.	Ch2 \$2.3; SLO (explain the allotropic forms of solids: diamond, graphite, fullerenes)
3	Which statement belongs to Bohr's model of the atom but NOT to the quantum mechanical model?	A. Electrons revolve around the nucleus in fixed circular orbits of definite energy. 1	Fixed circular orbits are Bohr's first postulate. The quantum mechanical model replaced exact orbits with probability regions called orbitals, treats electrons as wave-particles, and accepts Heisenberg's uncertainty principle.	Ch3 \$3.1; SLOs (atomic model as an aid; electrons as quantum particles, uncertainty principle)
4	When a nucleus emits a NEUTRON during radioactive decay, the atom changes into:	C. another isotope of the same element 1	Section 3.3: emitting a neutron changes only the neutron number, so the atom becomes another isotope. Emitting a PROTON changes the proton number and makes it another element (the twin case).	Ch3 \$3.3; SLO (radioactivity can change the proton number and alter an atom's identity)
5	The amount of energy RELEASED when an electron is added to the valence shell of an isolated gaseous atom, forming a uni-negative ion, is called:	A. electron affinity 1	Section 4.3.4 defines electron affinity as the energy released when an electron is added. Ionization energy is the opposite process: the energy REQUIRED to remove the outermost electron.	Ch4 \$4.3.4; SLO (identify trends ... including electron affinity, ionization energy)
6	The transition metal used as a catalyst in the manufacture of margarine is:	B. nickel 1	Section 4.5 lists nickel for margarine, iron for the Haber process, and platinum and palladium for catalytic converters (platinum also for the Contact process).	Ch4 \$4.5; SLO (transition elements act as catalysts: Haber process, catalytic converters, Contact process, margarine)
7	Silicon dioxide (SiO ₂) is very hard and difficult to melt because it has:	D. a giant covalent structure in which each atom is joined to several others by strong covalent bonds 1	Section 5.5 contrasts SiO ₂ (giant covalent structure, hard, difficult to melt) with CO ₂ (simple molecules, weak forces between them, low melting point).	Ch5 \$5.5; SLO (explain the properties of compounds in terms of bonding and structure)
8	In the formula (NH ₄) ₂ SO ₄ , the subscript 2 written outside the bracket shows that one formula unit contains:	B. two ammonium ions 1	A subscript outside a bracket multiplies the whole polyatomic ion. Two NH ₄ ⁺ ions (total +2) balance one SO ₄ ²⁻ ion. The formula contains eight hydrogen atoms, not two.	Ch6 \$6.3 (ion table incl. ammonium and sulfate); Ch7 SLO (derive the formula of ionic compounds from ionic charges)

9	The removal of hydrogen from a substance is called:	C. oxidation 1	Chapter 7 Key Points: oxidation is the gain of oxygen, the LOSS of hydrogen, or the loss of electrons. Reduction is the twin: loss of oxygen, gain of hydrogen, gain of electrons.	Ch7 Key Points; SLO (define redox reactions in terms of transfer of oxygen, hydrogen, electrons)
10	When a few drops of water are added to blue anhydrous cobalt(II) chloride, it turns:	A. pink 1	Section 9.3: hydrated cobalt(II) chloride is a pink solid and anhydrous cobalt(II) chloride is blue. White is the colour of anhydrous copper(II) sulphate, the twin experiment.	Ch9 §9.3; SLO (addition of water to anhydrous copper(II) sulphate and cobalt(II) chloride)

SECTION B — Short Questions (6 × 4 = 24 Marks)

Q.2 (i) Saturated / unsaturated solutions and the effect of temperature (data table) 4

(a) A **saturated solution** is a solution that contains the **maximum amount of solute** that can dissolve in it **at a given temperature**. (Undissolved solute is in dynamic equilibrium with dissolved solute.) 1

(b) An **unsaturated solution** contains **less solute than is required to saturate it** at that temperature, so it can still dissolve more. 1

(c) NH_4Cl : solubility **increases** with temperature, from 37.5 g at 20 °C to 77 g at 100 °C. 1

$\text{Ca}(\text{OH})_2$: solubility **decreases** with temperature, from 0.173 g at 20 °C to 0.066 g at 100 °C. 1

Marking note: each half of (c) needs the direction AND a figure from the table. Direction with no data = ½.

Source: Ch2 §2.4.2, §2.4.3, §2.4.8 and Table 2.2; SLO (effect of temperature on solubility; unsaturated and saturated solutions).

Q.2 (ii) Isotopes of bromine 4

(a) Relative atomic mass = $[(79 \times 50.5) + (81 \times 49.5)] \div 100$ 1

= $(3989.5 + 4009.5) \div 100 = 7999 \div 100 = 79.99 \approx 80.0$ 1

(b) ^{81}Br : protons = 35, electrons = 35, neutrons = $81 - 35 = 46$. 1

(c) Both isotopes have the **same number of protons and electrons** (the same electronic configuration). Chemical properties depend on the electrons, so isotopes are chemically alike; they differ only in neutrons, which affects mass, not chemistry. 1

Source: Ch3 §3.3, §3.5, §3.5.5; SLOs (calculate relative atomic mass from abundances; determine protons and neutrons; isotopes affect mass but not chemical properties).

Q.2 (iii) Period, group and ions from electronic configuration 4

(a) X ($1s^2 2s^2 2p^6 3s^2$): valence shell $n = 3$, so **Period 3**; last electron in an s sub-shell with 2 valence electrons, so **Group 2**. 1

Y ($\dots 3s^2 3p^4$): valence shell $n = 3$, so **Period 3**; p block with $2 + 4 = 6$ valence electrons, group = $6 + 10 =$ **Group 16**. 1

Rule reminder: period = n of the valence shell; group = valence electrons (s block) or valence electrons + 10 (p block).

Award each mark only if both period and group are right with a reason.

(b) X loses 2 electrons: X^{2+} . Y gains 2 electrons: Y^{2-} . 1

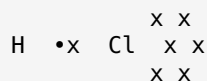
(c) Charges +2 and -2 balance 1 : 1, so the formula is **XY**. 1 (X is magnesium and Y is sulphur, MgS , but naming them is not required.)

Source: Ch4 §4.1.1 (period, group, block rules) and §4.2 (group number and ion charge); SLOs (identify group/period/block from configuration; relationship between group number and ion charge).

Q.2 (iv) HCl: dot-and-cross and polarity 4

(a) 2

H has 1 electron (•). Cl has 7 outer electrons (x).



•x = the ONE shared (bond) pair between H and Cl
x x = three lone pairs on Cl (6 electrons)

H now has 2 electrons (duplet, like He).
Cl now has 8 electrons (octet, like Ar).

1 mark: one shared pair (one • from H and one x from Cl) between the two atoms. 1 mark: the other six chlorine electrons shown as three lone pairs, giving Cl eight and H two.

(b) The bond is **polar covalent**. 1 H and Cl are **different atoms**; chlorine is **more electronegative**, so it pulls the shared electron pair towards itself, becoming partially negative (δ^-) while hydrogen becomes partially positive (δ^+).

1

Source: Ch5 §5.3.2 and §5.3.3 (H–Cl is the book's example of a polar covalent bond); SLO (draw the structure of simple molecules including HCl).

Q.2 (v) Redox: $2\text{NH}_3 + 3\text{CuO} \rightarrow 3\text{Cu} + \text{N}_2 + 3\text{H}_2\text{O}$ 4

(a) CuO: Cu + (–2) = 0, so Cu = +2. In Cu (free element): 0. 1

(b) NH_3 : N + 3(+1) = 0, so N = –3. In N_2 (free element): 0. 1

(c) **Nitrogen is oxidized** (–3 → 0, oxidation number increases). **Copper is reduced** (+2 → 0, oxidation number decreases). 1

(d) Oxidizing agent = **CuO** (it contains the copper that is reduced). Reducing agent = **NH_3** (it contains the nitrogen that is oxidized). 1

Marking note: the agent is the **whole molecule or formula unit**. Answers of “Cu” and “N” for (d) score 0. Inverted agents score 0.

Source: Ch7 §7.2.2 (oxidation number rules), §7.4.2 (oxidizing/reducing agents; whole molecule rule), Review Q6(e); SLOs (free element = 0; sum in neutral compound = 0; identify oxidizing and reducing agents).

Q.2 (vi) $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ in a sealed container 4

(a) Forward: SO_2 and O_2 form SO_3 (left to right). Reverse: SO_3 decomposes into SO_2 and O_2 (right to left). 1

(b) At the start **no SO_3 is present**, so there is nothing to decompose. 1

(c) Any two: 2

- the rate of the forward reaction **equals** the rate of the reverse reaction;
- the concentrations of reactants and products **remain constant** (they are not equal, just unchanging);
- both reactions **continue** to happen, so the equilibrium is **dynamic**;
- this only happens in a **closed** system.

Source: Ch9 §9.1 (the SO_2 and O_2 sealed-container discussion and Fig. 9.1), §9.2; SLO (reversible reactions reach equilibrium in a closed system when forward and backward rates are equal).

SECTION C — Long Questions (2 × 8 = 16 Marks)

Q.3 (i) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}(\text{g})$ from bond energies 8

(a) ΔH = sum of bond dissociation energies of reactants – sum of bond formation energies of products

Bonds broken: 1 $\text{N}\equiv\text{N}$ + 1 $\text{O}=\text{O}$ = 945 + 498 = **1443 kJ** 1

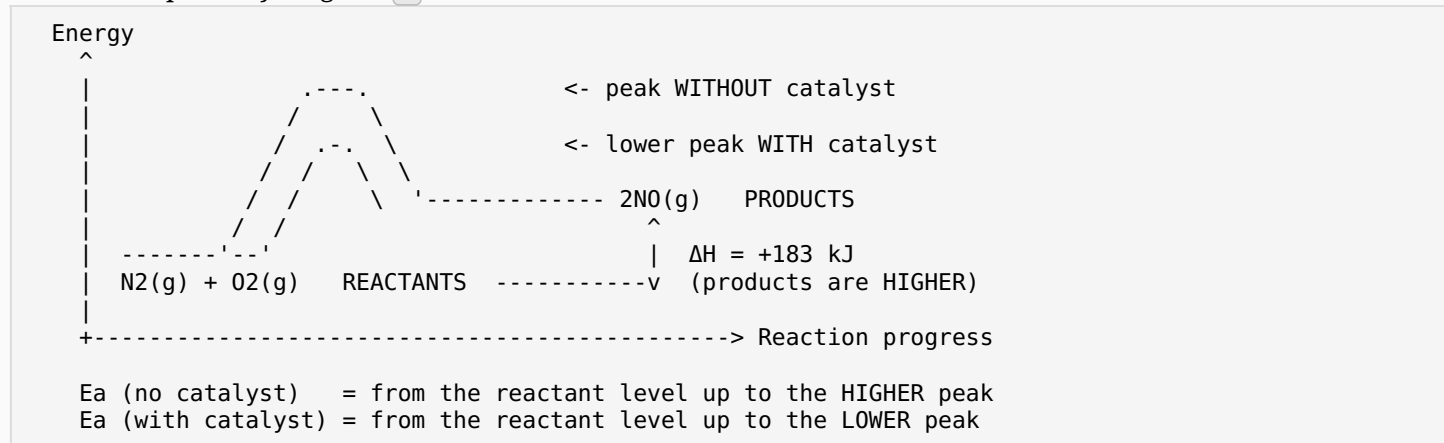
Bonds formed: 2 $\text{N}=\text{O}$ = 2 × 630 = **1260 kJ** 1

ΔH = 1443 – 1260 = **+183 kJ/mol** 1 (sign and unit required for this mark)

Note for checking: the textbook lists this reaction with $\Delta H^\circ = +180.5$ kJ; bond-energy values give an approximate answer, as Section 8.4 itself says.

(b) **Endothermic**, because ΔH is **positive**: more energy is absorbed breaking the bonds (1443 kJ) than is released forming new ones (1260 kJ). 1

(c) Reaction pathway diagram 3



1 mark: products drawn **higher** than reactants, with ΔH shown as the gap and marked positive.

1 mark: a single hump labelled activation energy (uncatalysed), measured from reactants to the peak.

1 mark: a **lower** hump labelled activation energy with catalyst, starting and ending at the **same** reactant and product levels.

(d) A catalyst **lowers the activation energy** (provides a new mechanism, so the reaction is faster), but it does **not change the enthalpy change ΔH** of the reaction. 1

Source: Ch8 §8.2.2 (this reaction printed as endothermic), §8.3, §8.4 (ΔH formula), §8.5, §8.6 and Fig. 8.3; SLOs (calculate ΔH from bond energies; sign of ΔH ; draw, label and interpret a pathway diagram with catalysed and uncatalysed E_a ; catalysts change the pathway).

Q.3 (ii) Hydrocarbon P: 2.1 g contains 1.8 g C and 0.3 g H; $M_r = 42$ 8

(a) Moles of C = $1.8 \div 12 = 0.15$ mol; moles of H = $0.3 \div 1 = 0.30$ mol. 1

Ratio C : H = 0.15 : 0.30 = 1 : 2, so the **empirical formula is CH_2** . 1

(b) Empirical formula mass = 12 + 2 = 14; $n = 42 \div 14 = 3$; **molecular formula = C_3H_6** . 1

(c) Moles = mass $\div$ molar mass = $8.4 \div 42 = 0.2$ mol. 1

(d) Molecules = $0.2 \times 6.022 \times 10^{23} = 1.204 \times 10^{23}$ molecules. 1

(e) **$2\text{C}_3\text{H}_6(\text{g}) + 9\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l})$**

Check: C 6 = 6; H 12 = 12; O 18 = 12 + 6. 1 for correct balancing, 1 for all four state symbols correct.

(f) Mass of one molecule = molar mass $\div N_A = 42 \div (6.022 \times 10^{23}) = 6.97 \times 10^{-23}$ g. 1

Marking note: carry her own values forward. If (a)/(b) are wrong but later parts correctly use her formula and molar mass, award those parts. Part (e) is the “answer every instruction” check: balancing without state symbols is 1 of 2.

Source: Ch6 §6.1 (empirical and molecular formula), §6.4, §6.5.1–6.5.2 (moles, mass, particles), §6.6 (balancing with state symbols); SLOs (define EF and MF; use amount = mass/molar mass for moles, mass, molar mass and number of particles; construct equations including state symbols).

Answer-letter balance: 3 A, 3 B, 2 C, 2 D (sequence BDACABDBCA).

— End of Answer Key —